

If x goes from $\pi/2 + i\infty$ to $\pi/2$, then $f(x)$ goes from 0 to i , and as x goes from $\pi/2$ to $-\pi/2$, then $f(x)$ travels from i to $-i$ on the half-circle. Finally, as x goes from $-\pi/2$ to $-\pi/2 + i\infty$, we see that $f(x)$ travels from $-i$ back to 0.

The mapping f is closely related to the inverse of the map in Example 5.

EXAMPLE 7. The function $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane (Exercise 5).

The boundary behavior of f is as follows. If x travels from 0 to 1, then $f(x)$ goes from ∞ to 1 on the real axis. If $z = e^{i\theta}$, then $f(z) = \cos \theta$ and as x travels from 1 to -1 along the unit half-circle in the upper half-plane, the $f(x)$ goes from 1 to -1 on the real segment. Finally, when x goes from -1 to 0, $f(x)$ goes from -1 to $-\infty$ along the real axis.

EXAMPLE 8. The map $f(z) = \sin z$ takes the upper half-plane conformally onto the half-strip $\{w = x + iy : -\pi/2 < x < \pi/2, y > 0\}$. To see this, note that if $\zeta = e^{iz}$, then

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = \frac{-1}{2} \left(i\zeta + \frac{1}{i\zeta} \right),$$

and therefore f is obtained first by applying the map in Example 6, then multiplying by i (that is, rotating by $\pi/2$), and finally applying the map in Example 7.

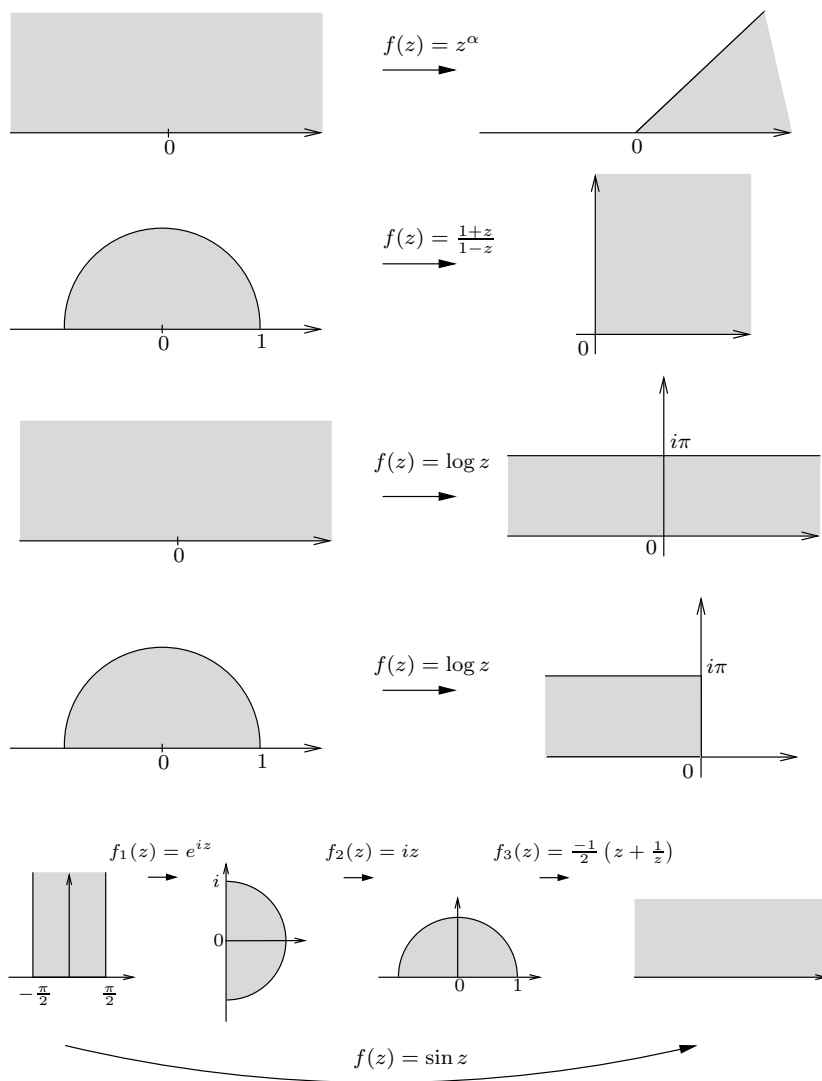
As x travels from $-\pi/2 + i\infty$ to $-\pi/2$, the point $f(x)$ goes from $-\infty$ to -1 . When x is real, between $-\pi/2$ and $\pi/2$, then $f(x)$ is also real between -1 and 1. Finally, if x goes from $\pi/2$ to $\pi/2 + i\infty$, then $f(x)$ travels from 1 to ∞ on the real axis.

1.3 The Dirichlet problem in a strip

The Dirichlet problem in the open set Ω consists of solving

$$(1) \quad \begin{cases} \Delta u = 0 & \text{in } \Omega, \\ u = f & \text{on } \partial\Omega, \end{cases}$$

where Δ denotes the Laplacian $\partial^2/\partial x^2 + \partial^2/\partial y^2$, and f is a given function on the boundary of Ω . In other words, we wish to find a harmonic function in Ω with prescribed boundary values f . This problem was already considered in Book I in the cases where Ω is the unit disc or the

**Figure 1.** Explicit conformal maps

upper half-plane, where it arose in the solution of the steady-state heat equation. In these specific examples, explicit solutions were obtained in terms of convolutions with the Poisson kernels.

Our goal here is to connect the Dirichlet problem with the conformal maps discussed so far. We begin by providing a formula for a solution to the problem (1) in the special case where Ω is a strip. In fact, this exam-

ple was studied in Problem 3 of Chapter 5, Book I, where the problem was solved using the Fourier transform. Here, we recover this solution using only conformal mappings and the known solution in the disc.

The first important fact that we use is that the composition of a harmonic function with a holomorphic function is still harmonic.

Lemma 1.3 *Let V and U be open sets in $\mathbb{C}$ and $F : V \rightarrow U$ a holomorphic function. If $u : U \rightarrow \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V .*

Proof. The thrust of the lemma is purely local, so we may assume that U is an open disc. We let G be a holomorphic function in U whose real part is u (such a G exists by Exercise 12 in Chapter 2, and is determined up to an additive constant). Let $H = G \circ F$ and note that $u \circ F$ is the real part of H . Hence $u \circ F$ is harmonic because H is holomorphic.

For an alternate (computational) proof of this lemma, see Exercise 6.

With this result in hand, we may now consider the problem (1) when Ω consists of the horizontal strip

$$\Omega = \{x + iy : x \in \mathbb{R}, 0 < y < 1\},$$

whose boundary is the union of the two horizontal lines $\mathbb{R}$ and $i + \mathbb{R}$. We express the boundary data as two functions f_0 and f_1 defined on $\mathbb{R}$, and ask for a solution $u(x, y)$ in Ω of $\Delta u = 0$ that satisfies

$$u(x, 0) = f_0(x) \quad \text{and} \quad u(x, 1) = f_1(x).$$

We shall assume that f_0 and f_1 are continuous and vanish at infinity, that is, that $\lim_{|x| \rightarrow \infty} f_j(x) = 0$ for $j = 0, 1$.

The method we shall follow consists of relocating the problem from the strip to the unit disc via a conformal map. In the disc the solution $\tilde{u}$ is then expressed in terms of a convolution with the Poisson kernel. Finally, $\tilde{u}$ is moved back to the strip using the inverse of the previous conformal map, thereby giving our final answer to the problem.

To achieve our goal, we introduce the mappings $F : \mathbb{D} \rightarrow \Omega$ and $G : \Omega \rightarrow \mathbb{D}$, that are defined by

$$F(w) = \frac{1}{\pi} \log \left(i \frac{1-w}{1+w} \right) \quad \text{and} \quad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}}.$$

These two functions, which are obtained from composing mappings from examples in the previous sections, are conformal and inverses to one

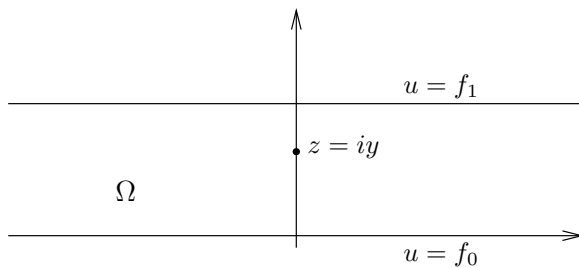


Figure 2. The Dirichlet problem in a strip

another. Tracing through the boundary behavior of F , we find that it maps the lower half-circle to the line $i + \mathbb{R}$, and the upper half-circle to $\mathbb{R}$. More precisely, as φ travels from $-\pi$ to 0 , then $F(e^{i\varphi})$ goes from $i + \infty$ to $i - \infty$, and as φ travels from 0 to π , then $F(e^{i\varphi})$ goes from $-\infty$ to ∞ on the real line.

With the behavior of F on the circle in mind, we define

$$\tilde{f}_1(\varphi) = f_1(F(e^{i\varphi}) - i) \quad \text{whenever } -\pi < \varphi < 0,$$

and

$$\tilde{f}_0(\varphi) = f_0(F(e^{i\varphi})) \quad \text{whenever } 0 < \varphi < \pi.$$

Then, since f_0 and f_1 vanish at infinity, the function $\tilde{f}$ that is equal to $\tilde{f}_1$ on the lower semi-circle, $\tilde{f}_0$ on the upper semi-circle, and 0 at the points $\varphi = \pm\pi, 0$, is continuous on the whole circle. The solution to the Dirichlet problem in the unit disc with boundary data $\tilde{f}$ is given by the Poisson integral³

$$\begin{aligned} \tilde{u}(w) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - \varphi) \tilde{f}(\varphi) d\varphi \\ &= \frac{1}{2\pi} \int_{-\pi}^0 P_r(\theta - \varphi) \tilde{f}_1(\varphi) d\varphi + \frac{1}{2\pi} \int_0^{\pi} P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi, \end{aligned}$$

where $w = re^{i\theta}$, and

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}$$

³We refer the reader to Chapter 2 in Book I for a detailed discussion of the Dirichlet problem in the disc and the Poisson integral formula. Also, the Poisson integral formula is deduced in Exercise 12 of Chapter 2 and Problem 2 in Chapter 3 of this book.

is the Poisson kernel. Lemma 1.3 guarantees that the function u , defined by

$$u(z) = \tilde{u}(G(z)),$$

is harmonic in the strip. Moreover, our construction also insures that u has the correct boundary values.

A formula for u in terms of f_0 and f_1 is first obtained at the points $z = iy$ with $0 < y < 1$. The appropriate change of variables (see Exercise 7) shows that if $re^{i\theta} = G(iy)$, then

$$\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi = \frac{\sin \pi y}{2} \int_{-\infty}^\infty \frac{f_0(t)}{\cosh \pi t - \cos \pi y} dt.$$

A similar calculation also establishes

$$\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_1(\varphi) d\varphi = \frac{\sin \pi y}{2} \int_{-\infty}^\infty \frac{f_1(t)}{\cosh \pi t + \cos \pi y} dt.$$

Adding these last two integrals provides a formula for $u(0, y)$. In general, we recall from Exercise 13 in Chapter 5 of Book I, that a solution to the Dirichlet problem in the strip vanishing at infinity is unique. Consequently, a translation of the boundary condition by x results in a translation of the solution by x as well. We may therefore apply the same argument to $f_0(x+t)$ and $f_1(x+t)$ (with x fixed), and a final change of variables shows that

$$u(x, y) = \frac{\sin \pi y}{2} \left(\int_{-\infty}^\infty \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y} dt + \int_{-\infty}^\infty \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y} dt \right),$$

which gives a solution to the Dirichlet problem in the strip. In particular, we find that the solution is given in terms of convolutions with the functions f_0 and f_1 . Also, note that at the mid-point of the strip ($y = 1/2$), the solution is given by integration with respect to the function $1/\cosh \pi t$; this function happens to be its own Fourier transform, as we saw in Example 3, Chapter 3.

Remarks about the Dirichlet problem

The example above leads us to envisage the solution of the more general Dirichlet problem for Ω (a suitable region), if we know a conformal map F from the disc $\mathbb{D}$ to Ω . That is, suppose we wish to solve (1), where f is an assigned continuous function and $\partial\Omega$ is the boundary of Ω . Assuming we have a conformal map F from $\mathbb{D}$ to Ω (that extends to a continuous

bijection of the boundary of the disc to the boundary of Ω), then $\tilde{f} = f \circ F$ is defined on the circle, and we can solve the Dirichlet problem for the disc with boundary data $\tilde{f}$. The solution is given by the Poisson integral formula

$$\tilde{u}(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) \tilde{f}(e^{i\varphi}) d\varphi,$$

where P_r is the Poisson kernel. Then, one can expect that the solution of the original problem is given by $u = \tilde{u} \circ F^{-1}$.

Success with this approach requires that we are able to resolve affirmatively two questions:

- Does there exist a conformal map $\Phi = F^{-1}$ from Ω to $\mathbb{D}$?
- If so, does this map extend to a continuous bijection from the boundary of Ω to the boundary of $\mathbb{D}$?

The first question, that of existence, is settled by the Riemann mapping theorem, which we prove in the next section. It is completely general (assuming only that Ω is a proper subset of $\mathbb{C}$ that is simply connected), and necessitates no regularity of the boundary of Ω . A positive answer to the second question requires some regularity of $\partial\Omega$. A particular case, when Ω is the interior of a polygon, is treated below in Section 4.3. (See Exercise 18 and Problem 6 for more general assertions.)

It is interesting to note that in Riemann's original approach to the mapping problem, the chain of implications was reversed: his idea was that the existence of the conformal map Φ from Ω to $\mathbb{D}$ is a consequence of the solvability of the Dirichlet problem in Ω . He argued as follows. Suppose we wish to find such a Φ , with the property that a given point $z_0 \in \Omega$ is mapped to 0. Then Φ must be of the form

$$\Phi(z) = (z - z_0)G(z),$$

where G is holomorphic and non-vanishing in Ω . Hence we can take

$$\Phi(z) = (z - z_0)e^{H(z)},$$

for suitable H . Now if $u(z)$ is the harmonic function given by $u = \operatorname{Re}(H)$, then the fact that $|\Phi(z)| = 1$ on $\partial\Omega$ means that u must satisfy the boundary condition $u(z) = \log(1/|z - z_0|)$ for $z \in \partial\Omega$. So if we can find such a solution u of the Dirichlet problem,⁴ we can construct H , and from this the mapping function Φ .

⁴The harmonic function $u(z)$ is also known as the Green's function with source z_0 for the region Ω .

However, there are several shortcomings to this method. First, one has to verify that Φ is a bijection. In addition, to succeed, this method requires some regularity of the boundary of Ω . Moreover, one is still faced with the question of solving the Dirichlet problem for Ω . At this stage Riemann proposed using the “Dirichlet principle.” But applying this idea involves difficulties that must be overcome.⁵

Nevertheless, using different methods, one can prove the existence of the mapping in the general case. This approach is carried out below in Section 3.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

The statement and proof of the Schwarz lemma are both simple, but the applications of this result are far-reaching. We recall that a rotation is a map of the form $z \mapsto cz$ with $|c| = 1$, namely $c = e^{i\theta}$, where $\theta \in \mathbb{R}$ is called the angle of rotation and is well-defined up to an integer multiple of 2π .

Lemma 2.1 *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then*

- (i) $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
- (ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
- (iii) $|f'(0)| \leq 1$, and if equality holds, then f is a rotation.

Proof. We first expand f in a power series centered at 0 and convergent in all of $\mathbb{D}$

$$f(z) = a_0 + a_1z + a_2z^2 + \cdots.$$

Since $f(0) = 0$ we have $a_0 = 0$, and therefore $f(z)/z$ is holomorphic in $\mathbb{D}$ (since it has a removable singularity at 0). If $|z| = r < 1$, then since $|f(z)| \leq 1$ we have

$$\left| \frac{f(z)}{z} \right| \leq \frac{1}{r},$$

and by the maximum modulus principle, we can conclude that this is true whenever $|z| \leq r$. Letting $r \rightarrow 1$ gives the first result.

For (ii), we see that $f(z)/z$ attains its maximum in the interior of $\mathbb{D}$ and must therefore be constant, say $f(z) = cz$. Evaluating this expression

⁵An implementation of Dirichlet’s principle in the present two-dimensional situation is taken up in Book III.